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Original article

Scapulothoracic muscle activity during elevation exercises measured with surface and fine wire EMG: A comparative study between patients with subacromial impingement syndrome and healthy controls

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ABSTRACT

Background: The quality of the scapular movement depends on the coordinated activity of the surrounding scapulothoracic muscles. Besides the well-known changes in Trapezius and Serratus Anterior (SA) activity in patients with subacromial impingement syndrome (SIS), no studies exist that have investigated the activity of the smaller less superficial muscles that attach on the scapula (Pectoralis Minor (Pm), the Levator Scapulae (LS) and the Rhomboid Major (RM)) in a population with SIS, despite the hypothesized importance of these muscles in shoulder function.
Objectives: To investigate if patients with shoulder impingement syndrome (SIS) show differences in deeper and superficial lying scapulothoracic muscle activity in comparison with a healthy control group during arm elevation tasks.
Study design: Controlled laboratory study.
Methods: Activity of the deeper lying (LS, Pm and RM) and superficial lying scapulothoracic muscles (Trapezius and SA) was investigated with fine-wire and surface electromyography (EMG) in 17 subjects with SIS and 20 healthy subjects while performing 3 elevation tasks: scaption, wall slide and elevation with external rotation. Possible differences between the groups were studied with a linear mixed model (factor "group" and "exercise").
Results: For the Pm only, a significant main effect for "Group" was found: during the elevation exercises, the Pm was significantly more active in the SIS group in comparison with the healthy controls.
Conclusion: Patients with SIS show significantly higher Pm activity during elevation tasks in comparison with healthy controls. This study supports the idea of a possible role of the Pm in SIS.
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1. Introduction

The scapula plays an important role in the function of the shoulder. During humeral elevation of the arm, a complex scapular movement of upward rotation, posterior tilt and external rotation is needed to create a stable base for the glenohumeral joint (Kibler and McMullen, 2003). The quality of this scapular movement depends on the coordinated activity of the surrounding superficial (Trapezius and Serratus anterior (SA)) and the deeper lying scapulothoracic muscles, such as Pectoralis minor (Pm), Levator Scapulae (LS) and Rhomboid Major (RM). A lack of activation or excessive activation of scapulothoracic muscles may impede optimal scapular movement. Hypothetically, excessive activation of the deeper lying muscles (Pm, LS and RM) may impede the warranted scapular movement that is necessary during humeral elevation. The LS is believed to elevate the scapula and to work together with the Rhomboids to retract and rotate the scapula downwards (Escamilla et al., 2009). The Pm is believed to move the scapula to protraction, downward rotation, anterior tilt and internal rotation (Oatis, 2004.). Normal upward rotation may be influenced by excessive activation or tension in the LS or RM (Behrsin and Maguire, 1986). Also, excessive activation of the Pm muscle may

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hinder normal posterior scapular tipping that is necessary during humeral elevation (Borstad and Ludewig, 2005).

A lack of upward rotation and anterior tilting during elevation of the arm has been related to shoulder impingement syndrome (SIS) (Ludewig and Cook, 2000; Struyf et al., 2011; Timmons et al., 2012). The term shoulder impingement was first introduced by Neer (1972), who described the phenomenon as a mechanical compression of the subacromial structures against the anterior undersurface of the acromion and coracoacromial ligament. In more recent literature, impingement has been described as a group of symptoms rather than a specific diagnosis, and has been considered to be an umbrella of a variety of shoulder conditions (Lewis, 2009). The shoulder impingement syndrome (SIS) symptoms are mostly present when the arm is elevated or when overhead activities are performed (Michener et al., 2004; Hung et al., 2010). To date, most studies that have investigated scapulothoracic muscle activity in patients with SIS have focused on the Trapezius and SA (Ludewig and Cook, 2000; Cools et al., 2004; Bandholm et al., 2006; Santos et al., 2007; Cools et al., 2007a; Moraes et al., 2008; Roy et al., 2008; Diederichsen et al., 2009; Lin et al., 2011). No studies exist that investigate the scapulothoracic activity of the smaller and less superficial muscles that attach on the scapula (such as the Pm, the LS and RM) in a population with shoulder pain, despite the hypothesized importance of these muscles in shoulder function (Cagnie et al., 2014; Cools et al., 2014).

Information from electromyographic (EMG) studies on the activity of the superficial muscles in patients with shoulder pain (Ludewig and Cook, 2000; Cools et al., 2007b; Reinold et al., 2009; Ellenbecker and Cools, 2010) has been a basis for recommendations for the choice of exercises during treatment for patients with shoulder symptoms related to scapulothoracic dysfunction. It is believed that performing exercises which address the appropriate muscles can improve the quality of the scapular movement and restore "normal" movement patterns. Although it is very important to know if patients with impingement symptoms show different activity of the deeper lying muscles, it has never been a topic of investigation. The activity of the deeper lying muscles has been studied by Castelein et al. (2016) during different commonly used rehabilitation exercises (scaption, towel wall slide, elevation with external rotation), showing different muscle activity patterns based on the specific modality of the exercise, however these investigations were performed on healthy subjects without shoulder pain. It would be interesting to know if patients with SIS would show differences during the performance of these elevation exercises, often used in clinical practice (Castelein et al., 2016).

Therefore, the objective of this study was to investigate whether the activity of the deeper lying (in particular LS, Pm and RM) and superficial lying muscles is different in a population with SIS compared to a healthy control group during various elevation tasks. This knowledge will aid clinicians in developing more targeted rehabilitation exercises.

2. Materials and methods

2.1. Subjects

Two groups of subjects were recruited: a group with shoulder impingement syndrome (SIS group, n = 17) and a matched control group without symptoms (healthy control group, n = 20). Subjects were recruited via advertisement from the local community and university. Written informed consent was obtained from all participants. The study was approved by the ethics committee of X. SIS was determined by history taking and confirmed by physical examination performed by an experienced musculoskeletal physical therapist. Patients were included in the SIS group if they reported chronic shoulder pain (>1 month during the last year, with a minimum pain intensity of 3/10 on the Numeric Rating Scale) in the anterior deltoid region of their dominant shoulder and if at least 3 of the following criteria were positive: (1) Positive Neer sign, (2) Positive Hawkins sign, (3) Positive Jobe's sign, (4) Painful Arc, and (5) Positive Resistance Test against External Rotation (Michener et al., 2009). Subjects had to be able to perform full ROM of humeral elevation in the scapular plane and this was tested by the investigator before the start of the study.

Exclusion criteria were shoulder surgery or dislocation, loss of range of motion (ROM) (i.e. capsular pattern in case of adhesive capsulitis: external rotation > abduction > internal rotation), positive spurling test, >2 cortisone injections, one cortisone injections within the last month, systemic diseases, current symptoms in the neck region, total rotator cuff rupture and upper limb training or overhead sports > 6 h/week. The in- and exclusion criteria were exclusively based upon clinical tests. Medical imaging was not a requirement for this study.

2.2. General design

EMG data was collected from 5 scapulothoracic muscles (Trapezius (UT, MT, LT) SA, LS, Pm, RM) on the dominant side of each subject during the performance of 3 different humeral elevation tasks in the scapular plane (Castelein et al., 2016): (1) scaption (elevation in the scapular plane), (2) towel wall slide and (3) elevation with external rotation component (with resistance from a Theraband®).

2.3. Test procedure

The experimental session began with a short warm-up procedure with multidirectional shoulder movements, followed by the performance of a set of five the maximum voluntary isometric contractions (MVIC) of the muscles of interest (Castelein et al., 2015), including:

1. "Abduction 90°" (sitting)
2. "Horizontal Abduction with external rotation" (prone lying)
3. "Arm raised above head in line with LT muscle fibers" (prone lying)
4. "Shoulder flexion 135°" (sitting)
5. "Arm raised above head in line with Pm muscle fibers" (supine lying)

MVIC test positions were taught to each subject by the same investigator, and sufficient practice was allowed before formal data collection. Manual pressure was always applied by the same investigator and strong and consistent verbal encouragement from the investigator was given during each MVIC to promote maximal effort. All MVICs were performed prior to the different elevation exercises, except for the MVIC "Arm raised above head in line with Pm muscle fibers". This MVIC was always performed at the end (after the exercises) to avoid pressure on the electrodes of the dorsal muscles (due to their contact with the examination table because of the supine position). Each MVIC test position was performed 3 times (each contraction lasted for 5 s-controlled by a metronome) with at least 30 s rest between the different repetitions. There was a rest period of at least 1 min between the different test positions.

In the second part of the investigation, the subject performed three elevation tasks: (1) elevation in the scapular plane, (2) towel slide against a wall and (3) elevation with external rotation of a Theraband® (Figs. 1–3). The exercises were performed in random order (simple randomization: envelopes containing the name of

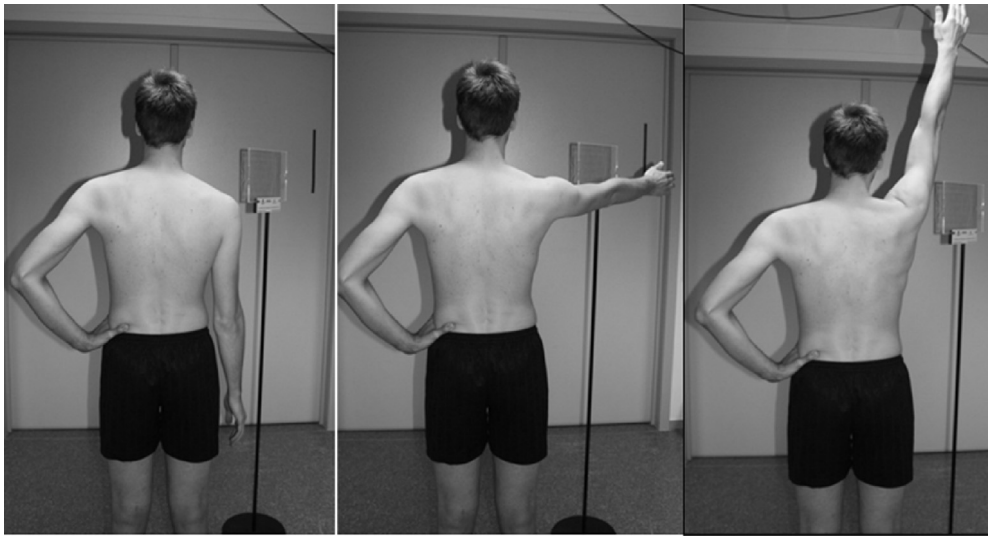

Fig. 1. Scaption. The subject performed elevation (full range of motion) with the dominant arm (thumb up) in the scapular plane (30°). A pole was used to guide the early phase of elevation in the scapular plane.

each exercise were shuffled for each participant and this sequence of exercises was allocated to that participant). Before data collection, the subject was given a visual demonstration of each exercise by the investigator. Each exercise consisted of an elevation phase of 4 s and a lowering phase of 4 s. For the exercise with the Theraband®, 2 s were added to induce tension and remove tension on the Theraband® before and after the elevation exercise. Only the influence during the elevation phase of this Theraband® (and not the tension of the glenohumeral external rotation position) on scapulothoracic muscle activity was of interest.

A metronome was used to control and standardize the speed of the movement. When the participants were able to perform the proper movement pattern and timing of the exercise, EMG data was collected from five repetitions of each exercise with 5 s of rest in between each trial. Between each exercise set, a break of 1.5 min was provided. Pain severity was asked during the performance of each of the elevation tasks (0: no pain – 10: worst possible pain).

2.4. Instrumentation

A TeleMyo 2400 G2 Telemetry System (Noraxon Inc., Scottsdale, AZ) was used to collect the EMG data. This study used a combination of surface and intramuscular electrodes. Bipolar circular surface electrodes (Ag/AgCl, Ambu® Blue Sensor P, Type N-00-S 30 × 22 mm, Ballerup, Denmark) were placed with a 1 cm interelectrode distance over the UT, LT and MT, according to the to the SENIAM (Surface Electromyography for the Non-Invasive Assessment of Muscles) Project Recommendations (Hermens et al., 1999). Electrodes for the SA were applied longitudinally (on the part where the muscle is most superficial): the first electrode anterior to the Latissimus Dorsi and the second electrode posterior to the Pectoralis Major (caudal from the axilla) (Lear and Gross, 1998; Decker et al., 1999; Ludewig et al., 2004; Maenhout et al., 2010). A reference electrode was placed over the processus spinosus of C7 vertebrae. Before surface electrode application, the skin surface was shaved, cleaned and scrubbed with alcohol to reduce impedance

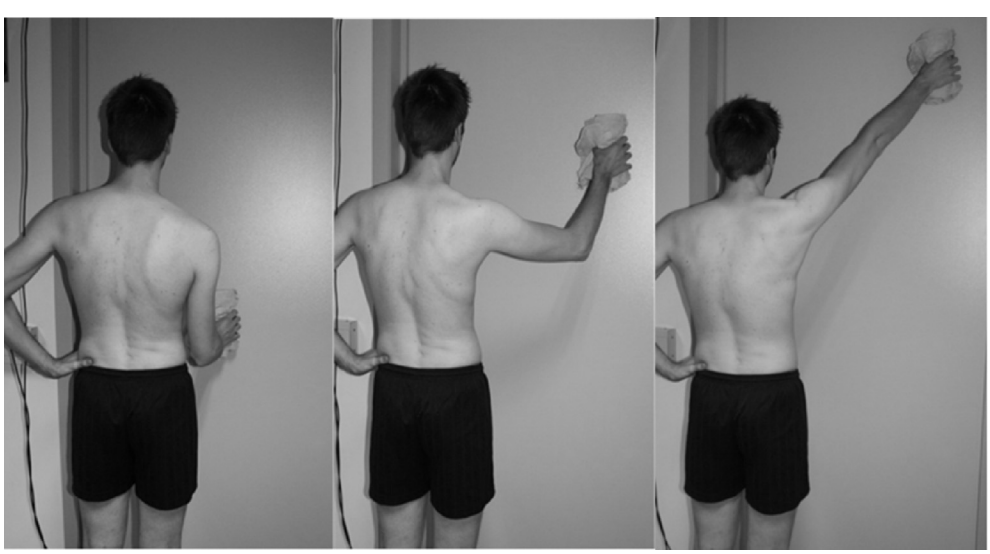

Fig. 2. Towel wall slide. For the starting position, the subject held a towel in the hand and put the hand against the wall with the elbow flexed 90°. The subject moved the towel up by sliding the arm against the wall until elbow was fully extended. This was performed in the scapular plane (30°). The distance between the wall and the subject was determined by the length of the forearm with the elbow in ninety degrees of flexion.

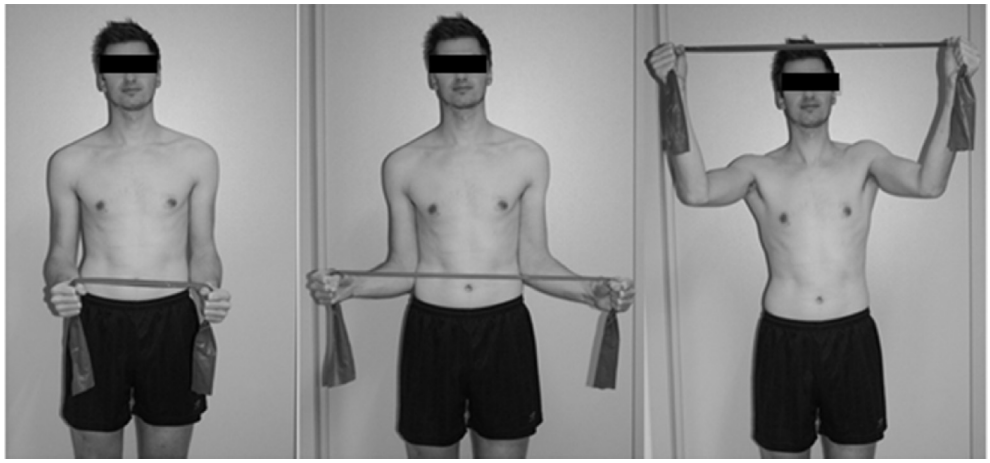

Fig. 3. Bilateral elevation with external rotation by holding a Theraband®. The subject took the Theraband® (color red was chosen to give a "medium" resistance) in both hands on two spots that the investigators marked on the Theraband®. The subject flexed the elbows 90° with the shoulder in a neutral position. The Theraband® was then brought to tension with 30° of external rotation in which the wrists remained in the neutral position. From this position an elevation of both arms was carried out up to 90° in the scapular plane while holding the tension of the Theraband®.

(<10 kOhm). Intramuscular fine-wire electrodes were used to measure the EMG activity of the LS, Pm and the RM. The paired hook fine-wire electrodes (Carefusion, Middleton, WI, USA – wire length 125 mm, stainless steel, insulated nickel alloy wire, first wire stripped 2 mm, second wire insulated for 3 mm and then stripped 2 mm) were inserted into the muscle belly (according to the locations described by Delagi et al. (1994)) using a single-use 25-gauge hypodermic needle. This was done using real-time ultrasound guidance, which has been shown to be an accurate and repeatable method of intramuscular electrode placement (Hodges et al., 1997). Correct electrode placement was checked by specific tasks that activated specific muscles. Following tasks were asked to the subject and the investigator checked the EMG signal of the corresponding muscle:

- For UT and SA: abduction 90° against resistance
- For Pm: protraction against resistance
- For LS: shrug against resistance
- For MT and RM: adduction of the scapula (retraction)
- For LT: arm raised above head in line with LT muscle fibers and resistance against further elevation

The surface and intramuscular electrodes were looped and taped on the skin to prevent them from being accidentally removed during the experiment and to minimize movement artifacts. The sampling rate was 3000 Hz. The device had a common mode rejection ratio of 100 dB. Gain was set at 1000 (baseline noise < 1 μV root-mean-square (RMS)).

2.5. Signal processing and data analysis

The Myoresearch 3.4 Master Edition Software Program was used for signal processing. The EMG signals were filtered with a high pass Butterworth filter of 20 Hz. Cardiac artifact reduction was performed, followed by full wave rectification and smoothing (root mean square, window 100 ms) of the signals. The windows of data were determined based on markers that were manually placed by the investigator during the testing. An average EMG value for each muscle and each participant was calculated for each exercise (8 s: 4 s concentric phase and 4 s eccentric phase). This average value was taken from the 3 intermediate repetitions, because the first and fifth repetitions were not used to control for distortion due to habituation or fatigue. These average EMG data were normalized and expressed as a percentage of their MVIC. For each MVIC, the average EMG value over a window of the peak 2.5 s of the 5 s was calculated. The average of the 3 trials was used for normalization (Stevens et al., 2007; Cools et al., 2007a; Maenhout et al., 2010; De Ridder et al., 2013). All five MVIC test positions were analyzed for each muscle (except the Pm activity was not analyzed during prone lying MVIC test positions and the other muscles' activity was not analyzed during supine lying MVIC test positions). The normalization value (100%) was the highest value for that muscle recorded during the 5 MVIC tests.

2.6. Statistical analysis

SPSS 22.0 was used for statistical analysis. Means ± standard deviations (SD) were calculated for the normalized EMG values (in % of MVIC) of the UT, MT, LT, SA, Pm, LS & RM for each exercise and for each group. For each muscle, a linear mixed model with 2 factors was performed: factor "exercise" (3 levels) and factor "group" (2 levels). The residuals of the linear mixed models were checked for normal distribution. Post hoc pairwise comparisons were performed using a Bonferroni correction. Only an interaction effect of "exercise × group" or a main effect of "group" was further interpreted. As we were only interested in the differences between groups, the main effect for the factor "exercise" was not further interpreted. An alpha level of 0.05 was applied to all the data in determining significant differences.

3. Results

3.1. Demographic characteristics of the participants

Thirty-seven female subjects were recruited. The SIS group consisted of 17 participants and the healthy control group of 20 participants. Table 1 shows the demographic characteristics of the participants. There were no significant differences in anthropometric data between the SIS group and the control group (independent sample t-test).

Table 1
Demographic characteristics of the participants. Values are expressed as means ± standard deviation.

	SIS group (n = 17)	Control group (n = 20)
Height, cm	167.4 ± 6.1	170.1 ± 5.9
Weight, cm	65.7 ± 9.4	62.9 ± 7.1
Age, y	30.1 ± 10.3	28.9 ± 11.5

3.2. Pain intensity during each of the elevation tasks

The Numeric Rating Score (mean ± SD) for pain during "Scaption" was 1.25 ± 1.76, for "towel wall slide" 1.18 ± 1.40 and for "Elevation with external rotation component" 2.6 ± 1.86.

3.3. EMG results

The mean EMG activity of each scapulothoracic muscle for both groups during the "Elevation exercises" is provided in Table 2. No significant "Exercise × Group" interaction was found. For the Pm only, a significant main effect for "Group" ($F = 5357$; $p = 0.023$) was found. Post-hoc analysis revealed that during the elevation exercises, the Pm was significantly more active in the SIS group in comparison with the healthy control group.

4. Discussion

The aim of the current study was to investigate if a population with SIS showed differences in deeper and superficial lying scapulothoracic muscle activity in comparison with a healthy control group during different arm elevation tasks. This is the first study that investigated the deeper lying scapulothoracic muscle activity in a population with SIS. The EMG data showed significantly higher Pm activity during all elevation tasks in the population with SIS. For the other deeper lying (RM and LS) and for the superficial lying scapulothoracic muscles no significant differences were found between the two groups during these specific movement tasks.

This is the first study investigating Pm activity in patients with shoulder pain, so no other data exist to compare our results with. Despite this lack of science based information of Pm activity in patients with shoulder pain, several authors have already suggested the important role of the Pm in shoulder pain. Studies that have suggested the role of Pm in patients with shoulder pain focused on shortening and tension of the Pm (Borstad and Ludewig, 2005, 2006; Borstad, 2008; Muraki et al., 2009). A shortened Pm may have influence on the position of the scapula. As the function of the Pm is to perform scapular protraction, downward rotation and anterior tilting, this muscle is passively lengthened during elevation of the arm as the scapula goes into upward, posterior tilt and external rotation of the scapula (Borstad and Ludewig, 2005, 2006; Borstad, 2006; Muraki et al., 2009). When Pm lacks extensibility, it can lead to a decrease in scapular upward rotation and an increase in scapular anterior tilting and internal rotation (Borstad and Ludewig, 2005; Borstad, 2008; Muraki et al., 2009), which can lead to a reduction of the subacromial space and has been linked to impingement symptoms (Borstad and Ludewig, 2005; Tate et al., 2012). These current results of higher activity of Pm during elevation of the arm in patients with SIS (mean difference with control group: 4.04 ± 1.7% MVIC, CI [0.577–7.50]) might gain insight into the relationship between the Pm and SIS. It is speculated that repetitive overuse of the Pm may result in adaptive shortening and tension and can lead to a malaligned scapula as it pulls the scapula anteriorly. However, the cause–consequence relationship between the activity of Pm and impingement symptoms still remains unclear.

Table 2
EMG activity (%MVIC ± standard deviation) of each scapulothoracic muscle in each group (healthy controls versus SIS patients) during the various "Elevation Exercises". The significance level of the p-value is also displayed.

Muscle	Population	Scaption	Towel wall slide	Elevation with external rotation	p-Value
UT	Healthy controls	17.7 ± 5.5	14.2 ± 4.6	12.3 ± 4.1	Exercise × Group NS
	SIS patients	18.7 ± 7.0	14.0 ± 6.5	12.0 ± 8.6	Group NS
	Mean group difference	1.00 ± 2.00	−0.20 ± 2.01	−0.23 ± 2.04	
	(SIS-healthy)	(CI: −2.98/4.99)	(CI: −4.18/3.78)	(CI: −4.38/3.81)	
MT	Healthy controls	11.1 ± 4.5	7.4 ± 5.7	21.0 ± 11.9	Exercise × Group NS
	SIS patients	13.9 ± 8.9	7.5 ± 4.8	24.8 ± 11.5	Group NS
	Mean group difference	2.86 ± 2.86	0.10 ± 2.81	3.81 ± 2.81	
	(SIS-healthy)	(CI: −2.82/8.55)	(CI: −5.48/5.68)	(CI: −1.78/9.39)	
LT	Healthy controls	15.7 ± 5.3	9.1 ± 4.4	29.3 ± 11.6	Exercise × Group NS
	SIS patients	15.6 ± 7.0	8.4 ± 4.7	27.0 ± 11.3	Group NS
	Mean group difference	−0.11 ± 2.60	−0.61 ± 2.60	−2.29 ± 2.65	
	(SIS-healthy)	(CI: −5.26/5.05)	(CI: −5.77/4.55)	(CI: −7.54/2.96)	
SA	Healthy controls	28.7 ± 14.5	26.8 ± 11.9	20.8 ± 9.0	Exercise × Group NS
	SIS patients	25.7 ± 9.5	25.3 ± 11.0	19.2 ± 5.2	Group NS
	Mean group difference	−2.99 ± 3.55	−1.52 ± 3.55	−1.62 ± 3.65	
	(SIS-healthy)	(CI: −10.0/4.04)	(CI: −8.55/5.51)	(CI: −8.86/5.61)	
Pm	Healthy controls	9.9 ± 7.6	12.3 ± 9.6	9.0 ± 7.6	Exercise × Group NS
	SIS patients	13.0 ± 8.4	17.5 ± 12.0	12.8 ± 7.5	Group p = 0.023[a]
	Mean group difference	3.09 ± 3.00	5.21 ± 2.95	3.8 ± 3.2	[a]Overall Mean Group difference:
	(SIS-healthy)	(CI: −2.86/9.04)	(CI: −0.65/11.1)	(−2.35/10.0)	−2.05 ± 2.07
					CI: −0.15/2.06
LS	Healthy controls	17.1 ± 11.0	13.7 ± 9.7	22.1 ± 17.4	Exercise × Group NS
	SIS patients	18.1 ± 12.0	13.3 ± 7.2	24.7 ± 17.4	Group NS
	Mean group difference	0.99 ± 4.4	−0.40 ± 4.38	2.62 ± 4.38	
	(SIS-healthy)	(CI: −7.70/9.57)	(CI: −9.09/8.27)	(CI: −6.06/11.3)	
RM	Healthy controls	26.0 ± 17.8	10.9 ± 4.6	31.3 ± 14.0	Exercise × Group NS
	SIS patients	25.3 ± 14.6	11.0 ± 9.2	31.1 ± 18.4	Group NS
	Mean group difference	−0.70 ± 4.67	0.07 ± 4.81	−0.23 ± 4.75	
	(SIS-healthy)	(CI: −9.97 ± 8.6)	(CI: −9.47/9.61)	(CI: −9.66/9.19)	

[a] UT = Upper Trapezius, MT = Middle Trapezius, LT = Lower Trapezius, SA = Serratus Anterior, Pm = Pectoralis Minor, LS = Levator Scapulae, RM = Rhomboid Major, NS = non-significant p-value, CI = 95% Confidence Interval for Difference (CI: Lower bound/Upper bound), significant difference if $p < 0.05$.

In the current study, no differences were found in LS and RM activity between patients with SIS and healthy controls. No other studies have investigated LS or RM activity in patients with shoulder pain, so no data exist to compare our results with. Hypothetically, overuse of these muscles during elevation could lead to downward rotation/diminished upward rotation of the scapula and a higher risk to impinge the subacromial structures. Nevertheless, the results of this study do not support this hypothesis. Future research should further investigate the role of these muscles in relation to shoulder pain.

For superficial muscle activity (Trapezius and SA), no differences were found between the two groups during elevation exercises. Several studies have already investigated the superficial scapulothoracic muscle activity in patients with SIS during elevation exercises (Ludewig and Cook, 2000; Bandholm et al., 2006; Roy et al., 2008; Lin et al., 2011). Our results were similar to the results of Bandholm et al. (2006) and Roy et al. (2008). Bandholm et al. (2006) did not find differences in UT, LT and SA EMG activity between 9 patients with SIS and 9 healthy controls during isometric elevation. Similar, Roy et al. (2008) did not find differences in Trapezius and SA activity between 33 individuals with SIS and 20 healthy individuals while reaching towards targets located at 90° of arm elevation in frontal and scapular plane. In contrast with these results, two other studies (Ludewig and Cook, 2000; Lin et al., 2011) showed differences in scapulothoracic EMG activity between patients with SIS and healthy controls during elevation. Lin et al. (2011) measured the UT, LT and SA activity in 14 overhead athletes with SIS (7 athletes and 7 students) and 7 overhead-playing controls during elevation in the scapular plane. They showed higher UT activity in the 7 SIS overhead athletes in comparison with the controls. LT and SA activity were found in both athletes and students with SIS in comparison with the control group. However, Lin et al. (2011) evaluated overhead athletes, whereas our study excluded overhead athletes (>6 h/week overhead sports), so comparison is rather difficult. Ludewig and Cook (2000) investigated UT, LT and SA activity in 26 overhead workers with SIS and 26 overhead workers without complaints during humeral elevation in the scapular plane under 3 conditions: (1) no load, (2) 2.3 kg load, and (3) 4.6 kg load and in 3 phases of motion (31°–60°, 61°–90° and 91°–120°). UT EMG activity was higher in patients with SIS, but only during the final 2 phases in the 4.6-kg load condition. LT EMG activity was increased in the group with SIS as compared with the group without SIS in the final 2 phases. The SA muscle demonstrated decreased activity in the group with impingement across all loads and phases.

Methodological aspects related to the tasks (plane of elevation, degrees of elevation, load, etc), different normalization procedures and inclusion criteria of the population could explain these differences across studies.

Some limitations have to be taken into account when interpreting the results of this study. Our EMG data were normalized to data of the MVICs. Discomfort or pain might have interfered with the ability to produce an MVIC. However, for normalization, we used the highest value of EMG activity for that muscle out of a set of 5 MVICs (Castelein et al., 2015). We believe that this set, rather than one MVIC for one muscle limits the role of pain as substantial confounding factor on the EMG results. In addition, discomfort or pain during the performance of the exercises might also have interfered with activity levels and patterns. Nevertheless, relative low levels of pain were reported during the performance of the exercises. Moreover, exercises were randomized to avoid the influence of exacerbation of pain upon the EMG activity. The fact that the sample size, needed for this study, was not calculated a priori is also a limitation of this study. Nevertheless, a post-hoc power calculation analysis (GPower®) revealed a power of 0.9992285 (with the EMG data from the Pm), which is very good. While this study provided useful information regarding the muscles being activated during various exercises, it did not provide synchronized 3D scapular kinematic analysis with the EMG signal. Investigating scapular movements, along with muscle activity during exercises, would provide additional information. Another limitation of this study is the inter-subject variability of patterns of activity as well as levels of activity that might have occurred during the performance of the different exercises. However, an attempt was made to mimic the clinical situation therefore variation of the tasks without additional feedback equipment were controlled using good explanation, good demonstration and personal feedback. Also, prospective clinical studies are also needed to determine whether rehabilitation programs that pay attention to the Pm are more effective than the current standard methods of care.

5. Conclusion

This study supports the idea of a possible role of the Pm in patients with SIS. Significantly higher Pm activity was found in patients with SIS during all elevation tasks. For the other scapulothoracic muscles (UT, MT, LT, SA, LS and RM) no significant differences were found between the two groups. The results of this study enhance the insight in scapulothoracic muscle dysfunction.

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